Systems of Linear Equations (AA HL)

Section 1: Augmented Matrices and Gaussian Elimination

1 In row-reducing an augmented matrix, which row operation is NOT permitted?

a. Squaring all elements in a row

b. Swapping two rows

c. Multiplying a row by a non-zero scalar

d. Adding a multiple of one row to another row

2 An augmented matrix in row echelon form has final row $[0, 0, 0 \mid 5]$. What does this indicate about the system?

a. The system is inconsistent (has no solutions)

b. The system has infinitely many solutions

c. The system has a unique solution with $z = 5$

d. The system has trivial solutions only

3 An augmented matrix in row echelon form has final row $[0, 0, 0 \mid 0]$ with 2 leading pivots across 3 variables. What does this signify?

a. The system has infinitely many solutions parameterized by 1 free variable

b. The system has a unique solution

c. The system has no solutions

d. The determinant of the matrix is 1

Section 2: Geometric Interpretation of 3 x 3 Systems

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4 What is the geometric configuration of three linear equations in three variables that possess a unique solution?

a. Three planes intersecting at a single unique point

b. Three planes intersecting along a common line

c. Three parallel non-intersecting planes

d. Three planes forming a triangular prism

5 If three planes intersect along a single straight line, what type of system is this?

a. Consistent dependent system with infinitely many solutions

b. Inconsistent system with no solutions

c. Consistent independent system with 1 solution

d. Homogeneous system with trivial solution only

6 Three planes form the faces of an open triangular prism with no common line or point. How is this system classified?

a. Inconsistent (no solutions)

b. Infinitely many solutions

c. Unique solution

d. Singular dependent

Section 3: Determinants and Cramer's Rule

7 Evaluate the 2×2 determinant $\det[[4, -3], [2, 5]]$.

a. 26

b. 14

c. -26

d. 20

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8 Under what condition does a 3×3 system $A\mathbf{x} = \mathbf{b}$ have a unique solution?

a. $\det(A) \neq 0$

b. $\det(A) = 0$

c. $\mathbf{b} = \mathbf{0}$

d. $\text{Tr}(A) > 0$

9 If $\det(A) = 0$ for a system $A\mathbf{x} = \mathbf{b}$, what can be concluded?

a. The system either has no solutions or infinitely many solutions

b. The system must have no solutions

c. The system must have infinitely many solutions

d. The system has exactly 2 solutions

Section 4: Parameterized Solutions and Free Variables

10 Solve the system: $x + y + z = 6$ and $2x - y + z = 3$. Parameterize the solution line using $z = \lambda$.

a. $x = 3 - \frac{2}{3}\lambda, y = 3 - \frac{1}{3}\lambda, z = \lambda$

b. $x = 3 + \lambda, y = 3 - \lambda, z = \lambda$

c. $x = 9 - 2\lambda, y = \lambda, z = \lambda$

d. $x = 1, y = 2, z = 3$

11 For what value of k does the system $x + 2y = 4$ and $3x + 6y = k$ have infinitely many solutions?

a. $k = 12$

b. $k = 4$

c. $k = 0$

d. $k = 6$

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12 If the homogeneous system $A\mathbf{x} = \mathbf{0}$ has non-trivial solutions (solutions other than $\mathbf{x} = \mathbf{0}$), what must be true about $\det(A)$?

a. $\det(A) = 0$

b. $\det(A) > 0$

c. $\det(A) = 1$

d. $\det(A) < 0$

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